PYTHON CODING LESSON & WORKSHEET

Python Project: News Headline Tracker

Learn how to store, search, filter, count, and export headlines using Python Lists, Loops, Conditionals, and Files!

🕵️ Today's Mission: Become a News Detective!

Imagine you are a news detective! Your mission is to write a step-by-step Python program that scans daily news headlines, searches for keywords like **ISRO**, **Cricket**, or **Weather**, calculates statistics, and saves output reports!

1. How Our News Tracker Works

CONCEPT	REAL-WORLD ANALOGY	PYTHON ROLE
Python List	A newsbag holding newspaper clippings	Stores multiple daily news headlines as text items in one variable.
For Loop	Flipping through news channels sequentially	Iterates through the list, inspecting each headline one by one.
If Statement	A detective checking for specific clues	Evaluates if a keyword exists in a headline.
String .lower()	Ignoring upper/lowercase handwriting differences	Normalizes text so searches match regardless of letter casing.
List Append / Accumulator	Collecting matching clippings in a folder / tally counter	Stores matching headlines in a new list and counts total occurrences.
File Handling (open / write)	Filing an official printed report	Saves filtered headlines into a text file on disk.

2. Step-by-Step Code Walkthrough

Step 1: Storing News Headlines in a List

```python
# Define initial list of daily headlines
headlines = [
    "ISRO launches new earth observation satellite",
    "IPL cricket updates: Mumbai wins thriller match",
    "Delhi weather forecast: Heavy rain expected tomorrow",
    "Neeraj Chopra wins gold medal in javelin throw",
    "Dussehra celebrations begin with grand festival",
    "ISRO announces upcoming lunar rover mission",
    "Stock market reaches record high today"
]

print(f"Loaded {len(headlines)} headlines successfully.")
```

Step 2: Basic Interactive Keyword Search

```
# Get user search keyword
search_word = input("Enter keyword to search (e.g., Weather, ISRO): ")

print(f"
--- Search Results for '{search_word}' ---")
for headline in headlines:
    # Convert both search word and headline to lowercase for flexible matching
    if search_word.lower() in headline.lower():
        print(f"ALERT! Found match: '{headline}'")
```

Step 3: Counting Matches and Filtering into a New List

```
# Store matched items and keep count
matched_headlines = []
match_count = 0

keyword = "ISRO"

for headline in headlines:
    if keyword.lower() in headline.lower():
        matched_headlines.append(headline)
        match_count += 1

print(f"Total matches found for '{keyword}': {match_count}")
print("Filtered Headlines List:", matched_headlines)
```

Step 4: Advanced Filtering by Headline Length

```
# Print only long headlines (more than 45 characters)
print("
--- Headlines Longer Than 45 Characters ---")
for headline in headlines:
    if len(headline) > 45:
        print(f"- [{len(headline)} chars]: {headline}")
```

Step 5: Exporting Search Results to a File

```
# Save matching headlines to a text file
search_topic = "ISRO"
filename = "headline_report.txt"

with open(filename, "w") as file:
    file.write(f"=== NEWS DETECTIVE REPORT: {search_topic} ===

")
    for headline in headlines:
        if search_topic.lower() in headline.lower():
            file.write(f"- {headline}
")

print(f"Report exported successfully to {filename}!")
```

Step 6: Building a Complete Interactive Headline Tracker Program

```python
# Complete automated tracker script
def run_news_tracker():
    headlines = [
        "ISRO launches new earth observation satellite",
        "IPL cricket updates: Mumbai wins thriller match",
        "Delhi weather forecast: Heavy rain expected tomorrow",
        "Neeraj Chopra wins gold medal in javelin throw",
        "Dussehra celebrations begin with grand festival"
    ]

    query = input("Enter keyword to track: ").strip()
    matches = [h for h in headlines if query.lower() in h.lower()]

    print(f"
Found {len(matches)} headline(s) matching '{query}':")
    for idx, match in enumerate(matches, 1):
        print(f"  {idx}. {match}")

# Run the program
# run_news_tracker()
```

3. Bug-Buster Challenge: Common Pitfalls

- **Missing Colons:** Python requires a colon `:` at the end of `for` loops and `if` statements.
- **Case Sensitivity:** Searching for `"isro"` will NOT match `"ISRO"` unless you convert both strings using `.lower()`.
- **Indentation Error:** Code inside loops and `if` statements must be indented consistently (4 spaces).
- **IndexOutOfBounds Error:** Remember that Python list indexing starts at `0`, so the last element of a list with 5 items is at index 4.

4. Practice Questions for Students

Question 1: Multiple Choice

Which Python data structure is used to store multiple headlines together in a single variable?

☐ A) Integer

☐ B) List

☐ C) Boolean

☐ D) Float

Question 2: Syntax Identification

Identify the syntax errors in the following snippet and write the corrected version:

```
for item in headlines
if "cricket" in item.lower()
print(item)
```

Question 3: Code Completion

Complete the missing parts (indicated by ___) to count matching headlines:

```
user_query = ___("Enter topic: ")
match_count = ___
for headline in headlines:
    if user_query.lower() ___ headline.lower():
        match_count = match_count + ___
print("Total matches found:", match_count)
```

Question 4: Mini Coding Task (Filtering by Length)

Write a Python loop that iterates through the `headlines` list and prints only headlines that contain more than 20 characters (Hint: use `len()`).

Question 5: File Output Task

Write a Python script that opens a file named `"results.txt"` and writes all headlines containing the keyword `"weather"` into it line by line.